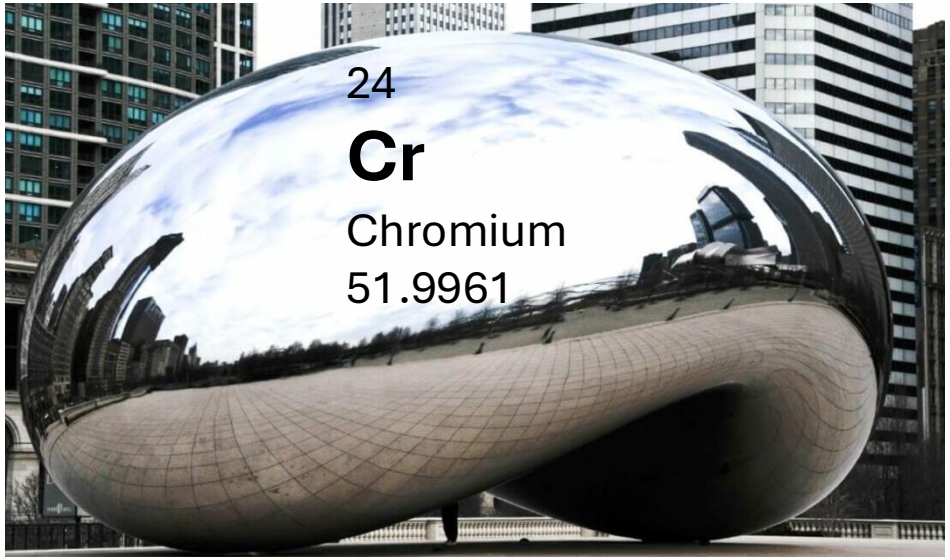
Environmental Chemistry and Community Action

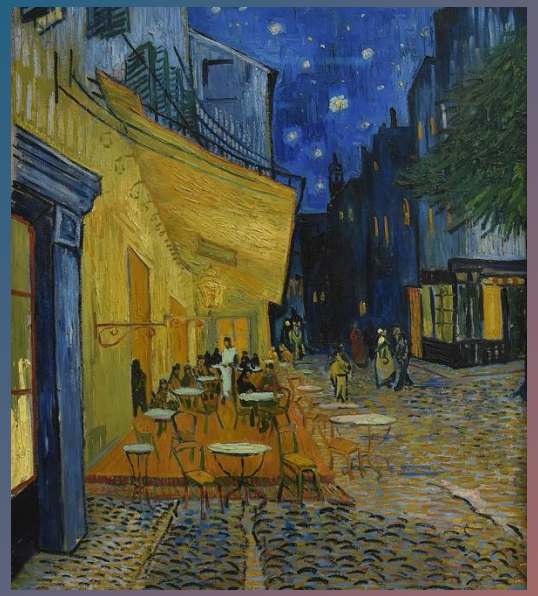
Chromium Basics

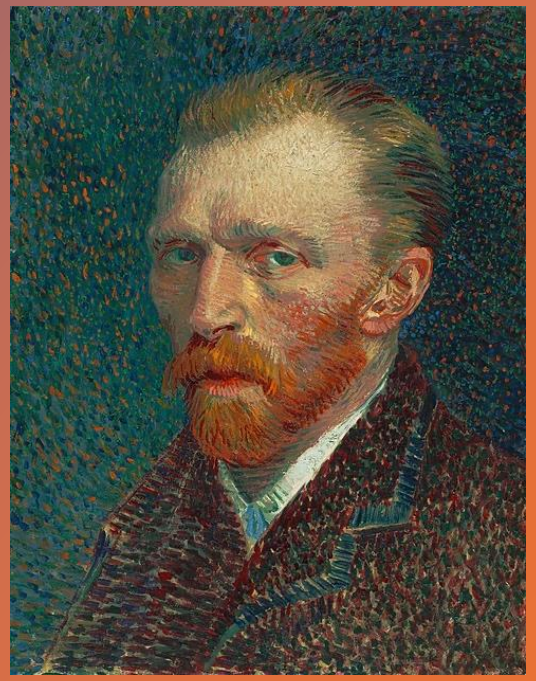
- Chromium is a naturally occurring metallic element with **Atomic Number 24**.
- It is **hard, steel-gray, and brittle** in appearance.
- Found in **rocks, soil, and volcanic dust**.
- Highly **resistant to corrosion and oxidation**.
- Essential for **stainless steel production and metal plating**.
- Exists mainly in two forms:
- **Trivalent Chromium (Cr^{3+})** – an essential nutrient that supports metabolism.
- **Hexavalent Chromium (Cr^{6+})** – a toxic industrial byproduct.



Chromium Basics

- Oxidation States of Chromium
  - Chromium III (trivalent): Essential nutrient, stable, less toxic
  - Chromium VI (hexavalent): Highly toxic, carcinogenic, oxidizing agent
- Chemical Forms of Chromium VI
  - Cr^{6+} exists as chromate (CrO_4^{2-})
  - Or dichromate ($Cr_2O_7^{2-}$)





Health & Environmental Impacts

- Known carcinogen (lung, nasal, sinus cancers)
- Causes skin ulcers, respiratory issues, kidney & liver damage
- Highly soluble → groundwater contamination risk
- Bioaccumulation potential

WATER QUALITY TESTING

Types of Water & How to Test Water

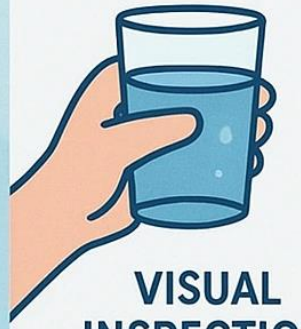
TYPES OF WATER

DRINKING WATER

SURFACE WATER

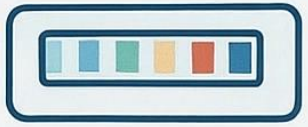
GROUND-WATER

HOW TO TEST WATER QUALITY



VISUAL INSPECTION

pH TEST



TEST STRIPS

LAB TEST

Guided Discussion

- Why would a power company use chromium VI in cooling towers?
- What made this contamination particularly dangerous?
- How does groundwater contamination spread?

Case study analysis

In your group, answer the assigned group questions and be ready to present to the rest of the class.

1. **The Science Group:** How did chromium VI enter the water supply? What concentrations are dangerous? What are EPA standards?
2. **The Community Impact Group:** Who was affected? Why did it take so long to discover? What is environmental justice?
3. **The Corporate Responsibility Group:** What are a company's obligations? How do regulations work? What is the precautionary principle?
4. **The Solutions Group:** How can such contamination be prevented? What monitoring systems exist? How do we remediate contaminated sites?